The Reach of Suction-Triggered Irrigation and the Marginal Fate of Irrigation Water: Closed Forms from the van Genuchten–Mualem Parameters

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Abstract

The van Genuchten–Mualem functions describe how much water a soil holds and how fast it conducts at any suction, and suction triggers decide when most sensor-scheduled fields are irrigated. Between the two lies a question that, to the author's knowledge, has no closed-form answer: which part of the retention curve can a crop actually reach at the sensor, so that a trigger set beyond it never fires? A second question, studied case by case in numerical work on pulsed and subsurface drip irrigation, is where each applied millimetre of water goes as the soil, the dose, the method and the season change. Both questions are answered here from the dry-end asymptote of the Mualem conductivity and from a large, openly archived ensemble of Daisy simulations covering ten soils and two decades of weather. Equating the relative conductivity at the reach to a single constant shared by all soils yields a closed-form reach that depends only on the van Genuchten shape parameters and the pore-connectivity exponent. It predicts the reach of soils left out of the fit to within about a factor of two, from sands, where the reach is short, to silty clay loams, where it lies far in the dry range. Commonly tabulated triggers lie beyond the reach of sandy soils: they never fire and the crop stays at its rainfed yield. Scaling each trigger by the reach places the irrigation responses of all soils on one axis. Less than half of each irrigation millimetre is transpired; most of the remainder evaporates and a smaller share drains below the root zone. The drained share rises with the dose towards a soil-specific plateau on a dose scale that is the same in every soil and close to one day of crop water use, so small doses trade drainage for evaporation, while overhead irrigation adds no transpiration at all. The closed form, the data, the analysis code and the software that applies them to any soil are open.

Keywords. van Genuchten–Mualem; irrigation trigger; matric-potential sensor; deep percolation; pulsed irrigation; marginal water productivity; Daisy model; decision support

1. Introduction

Since van Genuchten (1980), the retention curve of a soil and its unsaturated conductivity have been written with three shape parameters, α, n and the Mualem connectivity ℓ, and that closed form now sits inside almost every model of the unsaturated zone (Nielsen et al., 1986; Schaap et al., 2001; Šimůnek et al., 2016; van Genuchten and Nielsen, 1985). Irrigation is scheduled, on many fields with sensors, by a suction threshold: when a tensiometer or matric-potential sensor at a chosen depth reads more than a trigger h_{t}, water is applied (Allen et al., 1998). The triggers are taken from tables (commonly 300 to 600 cm for field crops), and the retention curve is used, if at all, to translate them into water contents.

Two questions lie between these lines of work. The first concerns the curve itself. The van Genuchten curve is defined for every suction, but a crop and a sensor visit only part of it. The wet end of that part has been given intrinsic definitions from the hydraulic parameters: field capacity as the state where internal drainage becomes negligible (Assouline and Or, 2014; Twarakavi et al., 2009). The dry end has not been given an equivalent closed form for control: which suction can roots actually impose at the depth of the sensor, so that a trigger set beyond it is never reached? Gardner (1960) showed that uptake becomes limited by the conductance of the soil around the roots long before the soil reaches the wilting point, and the same idea underlies modern analyses of stomatal response to soil drying (Carminati and Javaux, 2020). Yet no expression links that limit to α, n and ℓ in a form that an irrigation table could use.

The second question concerns the water once applied. In a series of numerical studies, Elmaloglou and Diamantopoulos (2007, 2008, 2009a,b, 2010) showed that pulsing, discharge, spacing, hysteresis and the hydraulic properties of the soil all change the wetting pattern and the losses below the root zone under drip; Diamantopoulos and Durner (2012) reviewed how flow departs from equilibrium; Weber et al. (2019) and Iden et al. (2021) extended the hydraulic functions to the dry range where films and vapour carry the flux; and Kumar et al. (2022) showed that site-specific schedules need site-specific hydraulic properties. Across that body of work, the share of an applied millimetre that leaves as transpiration, as evaporation or as deep percolation has been answered case by case, for one soil, one method and one dose at a time. How the shares depend jointly on the soil, the dose, the method and the season, and whether any of them follows a simple law, has not been mapped.

This paper answers both questions with one theory and one data set. The theory is the dry-end

asymptote of the Mualem conductivity. The data set is an ensemble of 10 190 Daisy simulations (Hansen et al., 2012; Holbak et al., 2026) whose every output is openly archived, year by year, in the repository folder `Data/Comprehensive_Summary` (Table 1).

Contributions. (1) A closed form for the reach h_c of a suction trigger from α, n and ℓ (Eq. (4)), with one constant, tested out of sample against interval-censored reaches of ten soils. (2) A reach-scaled trigger h_t/h_c that places the irrigation responses of all soils on one axis. (3) The marginal fate of an irrigation millimetre, split into transpiration, soil evaporation, other evaporation, drainage and storage, for eight soils, ten doses, ten seasons and three methods. (4) A soil-independent dose scale of the drained share and its interpretation. (5) The open data, code and software, and a manual through which each number can be reproduced.

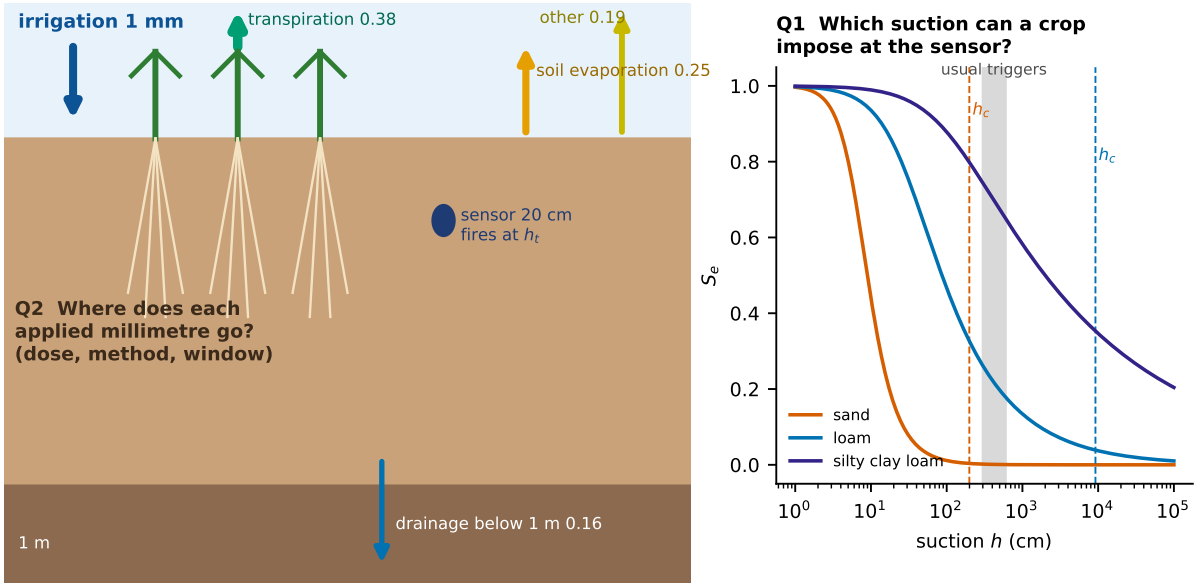


Figure 1: The two questions. Q1 (right): a trigger h_t fires only if the crop can dry the soil at the sensor to h_t; the reach h_c (dashed) is 199 cm in sand and 9×10^3 cm in loam, so the usual triggers (grey) lie beyond the reach of sand. Q2 (left): the fate of an irrigation millimetre, averaged over the 744 irrigated runs of experiment E10 (arrow widths to scale).

2. Theory

The van Genuchten–Mualem functions. With h the suction (cm), the effective saturation and the relative conductivity are (Mualem, 1976; van Genuchten, 1980)

$$S_e(h) = \left[1 + (\alpha h)^n\right]^{-m}, \qquad m = 1 - 1/n, \tag{1}$$

$$K_r(h) = S_e^{\ell}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2, \tag{2}$$

with $\ell = 0.5$ as usual. For $\alpha h \gg 1$, $S_e \simeq (\alpha h)^{-mn} = (\alpha h)^{-(n-1)}$ and $1-(1-S_e^{1/m})^m \simeq m\, S_e^{1/m}$, so

$$K_r(h) \;\simeq\; m^2\,(\alpha h)^{-[\,2n+\ell(n-1)\,]}, \qquad \alpha h \gg 1. \tag{3}$$

The exponent $2n + \ell(n-1)$ is 6.2 for the sand of this study and 2.6 for the silty clay loam: the conductivity of a coarse soil collapses over a narrow range of suction, that of a fine soil over many decades.

The reach of a trigger. Roots take water from a layer at a rate limited by the conductance of the soil around them, which scales with K (Carminati and Javaux, 2020; Gardner, 1960); Daisy computes uptake from a soil–root potential difference over such a conductance (Abrahamsen and Hansen, 2000; Hansen et al., 1990). As a layer dries, there is a suction beyond which the soil delivers too little water for the crop to dry it further within a season. If that point is reached at a fixed relative conductivity κ, Eq. (3) gives the reach of a trigger at the sensor:

$$\boxed{\,h_{\mathrm{c}} \;=\; \frac{1}{\alpha}\left(\frac{m^2}{\kappa}\right)^{1/[\,2n+\ell(n-1)\,]}\,} \tag{4}$$

A trigger $h_{\mathrm{t}} < h_{\mathrm{c}}$ can fire; a trigger $h_{\mathrm{t}} > h_{\mathrm{c}}$ is never reached and the crop is not irrigated, however dry it becomes. Equation (4) has one constant, κ, shared by all soils. It contains no saturated conductivity: two alternatives that do (an absolute conductivity $K(h_{\mathrm{c}}) = q_c$) and two that use only the retention curve (a fixed $S_e(h_{\mathrm{c}})$, or a power law in $n-1$) were fitted in the same way and are compared in Table 2. The reach-scaled trigger is $x = h_{\mathrm{t}}/h_{\mathrm{c}}$.

The marginal fate of irrigation. For a run with irrigation I and its rainfed twin on the same soil and weather (experiment E0), the change of each annual flux X per unit of irrigation is

$$\eta_X = \frac{X - X_0}{I}, \qquad X \in \{T, E_s, E_o, D\}, \qquad \eta_T + \eta_{E_s} + \eta_{E_o} + \eta_D + \eta_S = 1, \tag{5}$$

where T is transpiration, E_s soil evaporation, $E_o = ET - T - E_s$ the other evaporation (intercepted water, ponded water, litter and snow; the data folder shows that these close ET within 1.1 mm), D drainage at 1 m and η_S the change of storage and runoff, which closes the balance. Pooled over a set of runs, $\eta_X = \sum(X - X_0)/\sum I$. The share that drains is fitted, per soil, as

$$\eta_D(d) = \eta_\infty\big(1 - e^{-d/d_D}\big), \tag{6}$$

where d is the dose per event, η_∞ the plateau and d_D the dose scale.

Dynamic non-equilibrium: a criterion, not tested here. Under fast wetting and drying the water content lags the equilibrium of the retention curve, $\partial\theta/\partial t = [\theta_{\mathrm{eq}}(h) - \theta]/\tau$ (Diamantopoulos and Durner, 2012; Diamantopoulos et al., 2012; Hassanizadeh et al., 2002; Ross and Smettem, 2000). A sensor then reads a suction that is not the one of Eq. (1). For a schedule with interval t_i between applications, the dimensionless lag $\Pi_\tau = \tau/t_i$ says when this matters: for daily or multi-day scheduling ($t_i \geq 1$ d) and τ of minutes to hours, $\Pi_\tau \ll 1$ and the equilibrium reach of Eq. (4) applies; for pulses of minutes to hours, as in pulsed drip, $\Pi_\tau \sim 1$ and the sensor reading, the effective κ and the drained share become rate-dependent. Daisy solves the equilibrium Richards equation, so this criterion is stated for later testing and not evaluated here.

3. Data and methods

The simulation ensemble and its data. Daisy solves the one-dimensional Richards equation with the functions of Eqs. (1) and (2), coupled to a crop, a surface residue layer, an energy-based potential evapotranspiration and nitrogen turnover (Hansen et al., 1991, 2012; Holbak et al., 2026). The ensemble has 17 experiments (E0 to E16), ten soils from sand to silty clay loam, 20 years of Taastrup weather (1980–1999) and a cereal rotation; 10 041 of 10 190 runs are valid and 149 are solver failures, excluded (Table 1). The irrigation scheduler of Daisy applies a dose whenever the suction at a sensor depth exceeds a trigger. Every run is in the repository: the one-row-per-run summary (`00_summary_all_simulations.csv.gz`), every log of every run year by year (`E00_Baselines.csv.gz` to `E16_Edge_Cases.csv.gz`), and the summariser that made them, in `Data/Comprehensive_Summary`; the complete raw archive is on Zenodo (Halilu, 2026a). The soils are described by their effective van Genuchten parameters (Table 4) and, for eight of them, by literature saturated conductivities (Carsel and Parrish, 1988).

Table 1: The experiments of the analysis, from `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157

Experiments used. *E2* (197 runs) sets the trigger at 20 values from 50 to 10 000 cm on each soil (sensor at 20 cm, surface irrigation). *E3* (200 runs) moves the sensor from 5 to 100 cm at a 300 cm trigger. *E10* (954 runs) crosses the dose (1 to 30 mm per event, each applied in 12 h) with ten irrigation windows on each soil, at a 300 cm trigger and a 20 cm sensor. *E4* (96 runs) compares surface, overhead and subsurface application on each soil, and *E15* (1000 runs, coarse loam) crosses the method with the rate, the dose and the maximum rooting depth. *E0* gives the rainfed twin of every soil.

Reach, fit and validation. A trigger is said to fire on a soil if it brings at least 5 mm of irrigation a year. For each soil the reach is then known as an interval: above the largest firing trigger and below the smallest non-firing one, or only bounded from below when even 10 000 cm fires (five soils). The laws of Table 2 were fitted by minimising the squared distance, in $\log_{10}$, between the prediction and the interval (zero inside the interval), and validated by leaving each soil out in turn. The reach-scaled collapse is measured as the mean within-bin variance of the normalised irrigation (irrigation over its soil maximum) in ten quantile bins of $\log_{10} h_{\mathrm{t}}$ or of $\log_{10} x$, relative to the total variance.

Software. All analyses are in one script, `Manuscripts/reproducibility/analysis/p4_reach.py`, which reads the data folder and writes every number of this paper to `Manuscripts/results/p4_numbers.json` and every table and figure as a CSV file; the figures of the data are made by `data_overview.py`. The same quantities can be explored without code in SWAFL Desktop 1.1.0 (Section 5).

4. Results

A suction trigger has a reach, and the reach follows the curve. On every soil the irrigation falls as the trigger moves to drier values, but the fall ends in a cliff whose position differs by three orders of magnitude between soils (Fig. 3a). In sand no trigger drier than 100 cm fires; in loamy sand none drier than 200 cm; in sandy loam none drier than 800 cm; in loam and sandy clay loam none drier than 5 000 cm; the five other soils still irrigate at 10 000 cm. A 300 cm trigger is therefore blind on sand and loamy sand: the crop stays at its rainfed yield of 1.13 and 1.38 Mg ha^{-1}, against 3.04 and 4.54 Mg ha^{-1} reached with a 50 cm trigger (Fig. 3d). Moving the sensor from 5 to 100 cm does not cure the blindness: at a 300 cm trigger, sand receives no irrigation at any sensor depth and loamy sand at most 1.3 mm a year (E3).

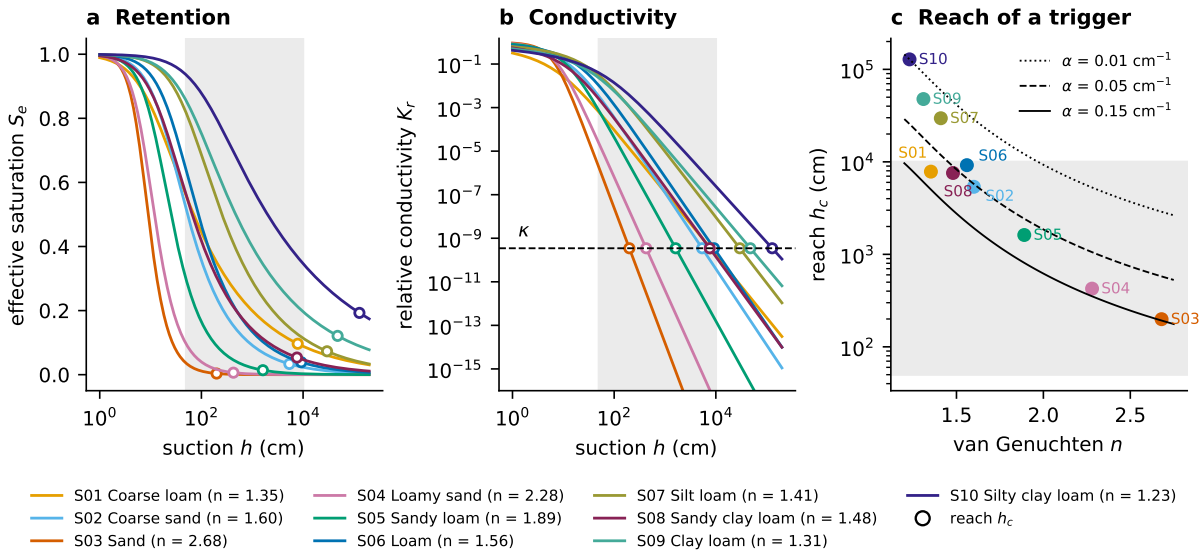


Figure 2: (a) Retention and (b) relative conductivity of the ten soils (Eqs. (1) and (2)); circles mark the reach of Eq. (4) and the dashed line the fitted $\kappa = 3.5 \times 10^{-10}$; grey: the range of triggers tested. (c) Reach against n for the ten soils, with Eq. (4) for three values of α.

Equation (4) places these cliffs with one constant (Figs. 2 and 3c). With $\kappa = 3.5 \times 10^{-10}$ the predicted reach is 199 cm for sand, 426 cm for loamy sand, 1 618 cm for sandy loam, 9 209 cm for loam and 7 580 cm for sandy clay loam, and above 5 000 cm for the five soils that still fire at 10 000 cm. Leaving each soil out, the prediction falls inside the observed interval for five soils and is never off by more than a factor of 2.3 (root-mean-square error 0.18 in $\log_{10}$; Table 2). The two-parameter power law in $n - 1$ is slightly closer (0.14) but has no derivation, and a single effective saturation at the reach does worse (0.23, a factor of 3.2). A threshold on the absolute conductivity $K(h_c)$ is not supported: with the literature K_s of eight soils the conductivity at the bracketing triggers spans $10^{-4.8}$ to $10^{-8.4}$ cm d^{-1}, more than three orders of magnitude, so the reach is set by the shape of the conductivity curve rather than by its level.

Table 2: Closed forms for the reach of a trigger, fitted to the interval-censored reaches of ten soils (E2) and validated leaving one soil out.

Law	Form	Fitted	LOSO RMSE ($\log_{10}$)	Inside (of 10)	Worst factor
Mualem asymptote, Eq. (4)	$K_r(h_c) = \kappa$	$\kappa = 3.5 \times 10^{-10}$	0.18	5	2.3
Power law	$\alpha h_c = A(n-1)^{-B}$	$A = 68$, $B = 3.1$	0.14	4	2.4
Fixed saturation	$S_e(h_c) = S_c$	$S_c = 0.032$	0.23	4	3.2
Absolute conductivity	$K(h_c) = q_c$	–	not supported (spans 3.6 decades)		

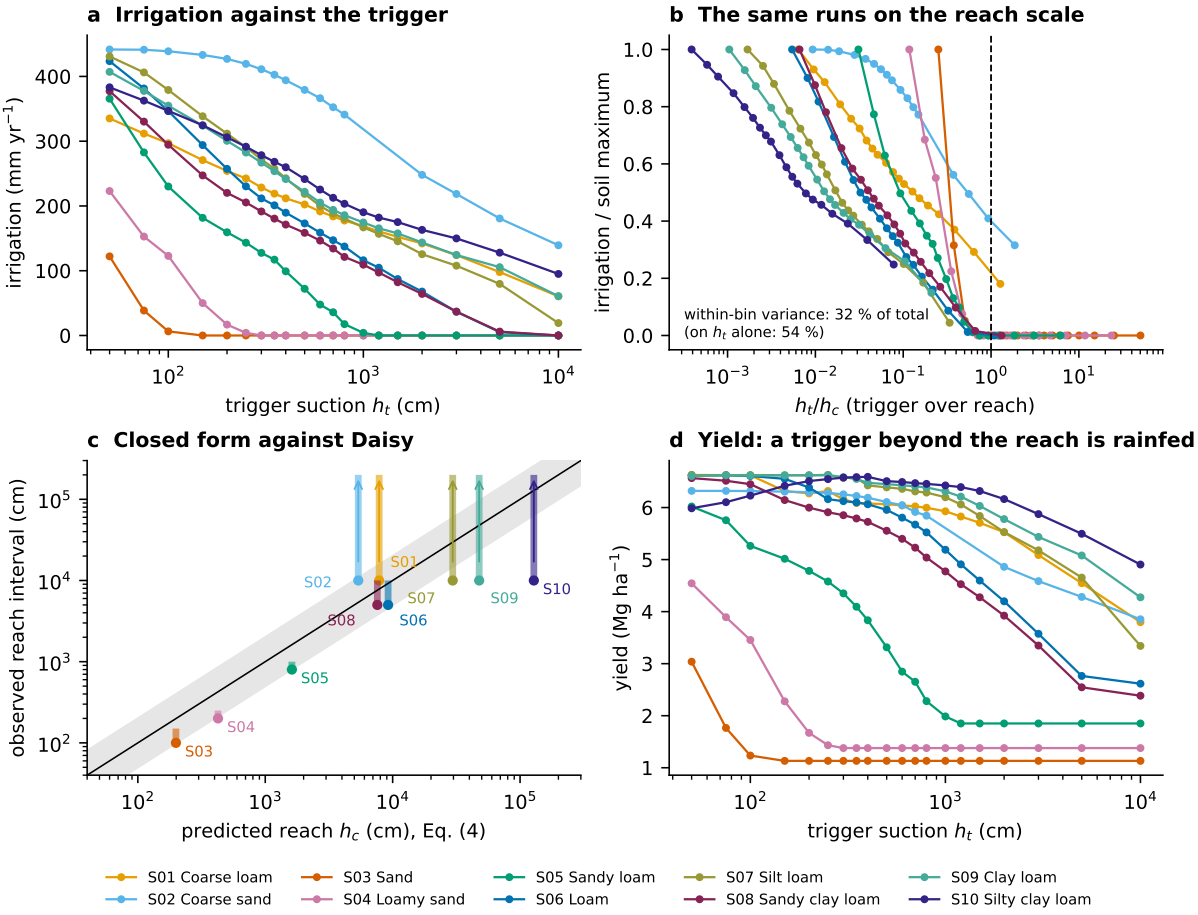


Figure 3: The reach of a suction trigger in experiment E2 (197 runs). (a) Irrigation against the trigger. (b) The same runs against the reach-scaled trigger h_t/h_c, irrigation divided by its soil maximum: every soil stops at $h_t/h_c \approx 0.5$ to 1. (c) Predicted reach (Eq. (4)) against the observed interval (bars; arrows: only a lower bound); grey band: within a factor of 2. (d) Yield against the trigger: beyond the reach the crop is rainfed.

One axis for all soils. On the reach-scaled axis the end of irrigation falls between $x = 0.5$ and 1 for every soil (Fig. 3b). The within-bin variance of the normalised irrigation drops from 54 % of the total on the raw trigger to 32 % on x (Spearman $\rho = -0.83$ between $\log x$ and the normalised irrigation). What is left is the slope of the approach, which is steep in coarse soils and shallow in fine ones; the reach sets where irrigation ends, not how it fades. A practical rule follows: a trigger should lie below about $h_c/2$, and the 300 cm of the tables satisfies this on eight of the ten soils but not on sand and loamy sand.

Where an irrigation millimetre goes. Pooled over the 744 irrigated runs of E10, a millimetre of irrigation became 0.38 mm of transpiration, 0.25 mm of soil evaporation, 0.19 mm of other evaporation, 0.16 mm of drainage below 1 m and 0.03 mm of storage and runoff (Fig. 4a). The transpired share is largest in sandy loam (0.64) and smallest in coarse sand (0.23), where 0.32

drains. In the five fine-textured soils the shares are close: 0.34 to 0.46 transpired, 0.15 drained, the rest evaporated.

The method changes the shares more than the soil (Fig. 4b). Under overhead irrigation the extra transpiration is nil or negative (-0.08 in E4, -0.01 in E15): the water intercepted by the canopy evaporates (0.79 and 0.56 of each millimetre) and lowers the transpiration it replaces. Subsurface application transpires most (0.51 in E4, 0.40 in E15) but also drains most (0.23 and 0.28), because it bypasses the surface and wets the profile from within. Surface application lies between. In E15 the extra drainage under subsurface application rises to 92 mm a year when the roots are limited to 30 cm, against 56 to 60 mm for any rooting depth of 45 cm or more.

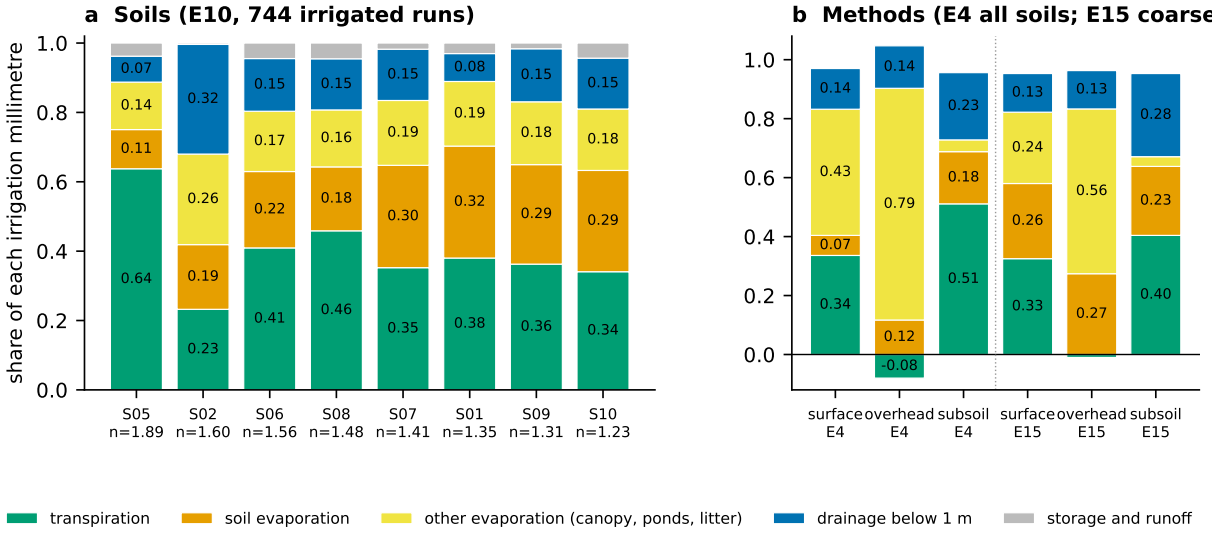


Figure 4: The fate of an irrigation millimetre, Eq. (5). (a) By soil (E10, 744 irrigated runs; sand and loamy sand are never irrigated at the 300 cm trigger of E10), ordered by n. (b) By method, on all soils (E4) and on coarse loam over all rates, doses and rooting depths (E15).

The dose: a soil-independent scale for drainage. The drained share rises with the dose and levels off (Fig. 5a). Equation (6) fits every soil (R^2 0.92 to 0.96 for seven soils, 0.66 for sandy loam) with a dose scale d_D of 3.2 to 5.2 mm, mean 4.1 ± 0.6 mm, which is not ordered by any property of the soil (Spearman $\rho = 0.07$ with n, -0.24 with the available water, 0.17 with the water released between 100 and 300 cm in the top 30 cm). The plateau is soil-specific: 0.10 to 0.20 in seven soils and 0.45 in coarse sand. A dose of 1 mm drains almost nothing (0.01 to 0.09), but 60 to 72 % of it evaporates (Fig. 5c); the transpired share rises with the dose up to 15 to 20 mm in most soils (Fig. 5b). The dose that maximises transpired minus drained water is 2 mm on six of eight soils.

A dose scale of about 4 mm, independent of the soil, is of the order of the daily water use of the crop in the irrigation season. The interpretation is that a dose smaller than one day of

evapotranspiration is used before it can move below the roots, whatever the soil, while the plateau, the share of a large dose that is lost, depends on how fast the soil drains once wetted, which is what the hydraulic parameters set. This is consistent with the pulsed-drip simulations of Elmaloglou and Diamantopoulos (2007, 2008), here generalised across ten soils and to the evaporation that small doses pay instead.

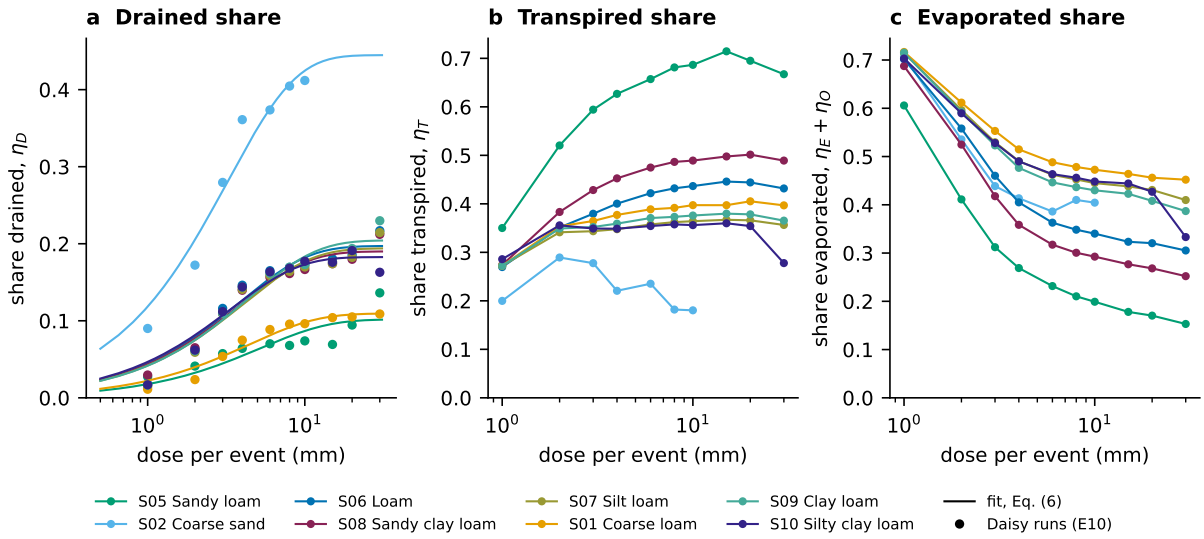


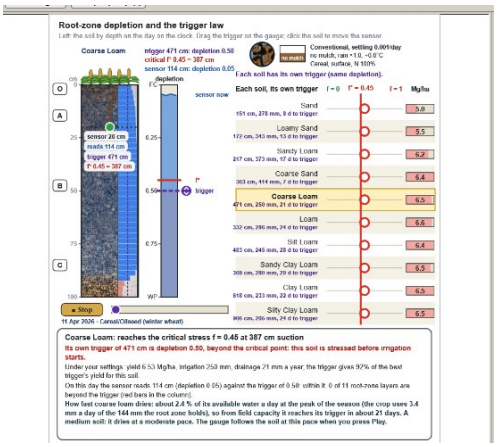
Figure 5: The dose (E10, all ten windows pooled). (a) Drained share with Eq. (6) per soil. (b) Transpired share. (c) Evaporated share (soil and other evaporation).

The season. The irrigation window shifts the fate as strongly as the dose. Irrigating only from April to June transpires 0.64 of each millimetre and drains 0.07; irrigating the whole season (April to October) transpires 0.34 and drains 0.21; irrigating only from July to mid-September transpires 0.23. Early water is used by a growing canopy; late water arrives when the cereal is ripening or harvested and is lost to evaporation and drainage.

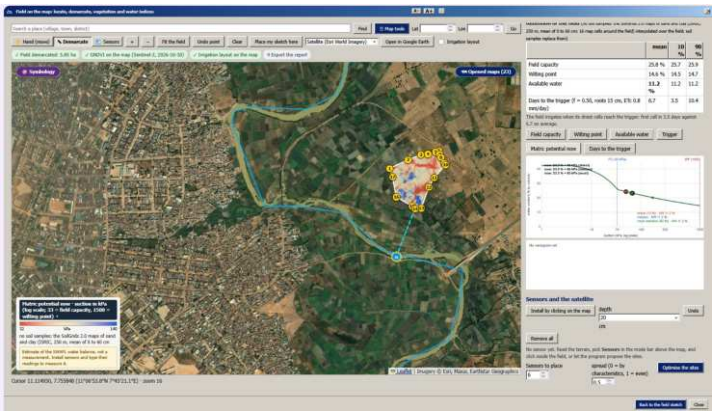
5. The answer in the software and in its manual

The results can be reproduced and extended without code. In SWAFL Desktop 1.1.0 the layer *Trigger law* draws the root-zone depletion on the day of the season, the depth of the sensor and the trigger of every soil on one gauge, and says, for the soil chosen, whether the trigger lies beyond the point the soil can reach (Fig. 6a; manual p. 56). On a real field near Zaria, Nigeria, the map of the field draws the matric potential cell by cell from the retention curves mapped from SoilGrids, so the reach can be read where a sensor is to be placed (Fig. 6b). The Daisy workbench runs new trigger, dose and method experiments on any soil, including soils scaled in α, n, K_s, θ_s and ℓ

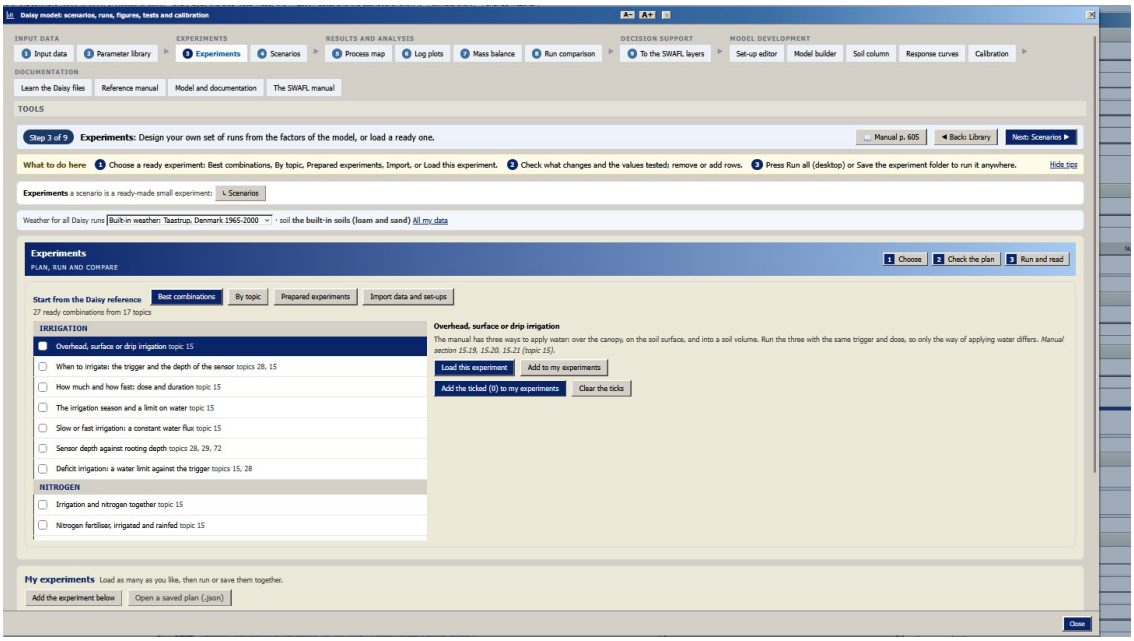
(Fig. 6c; manual p. 592). The 1155-page manual (Halilu, 2026b) documents each control, each output and the equations; the data of this paper are linked from it and from the repository.



(a) Trigger law layer: depletion, sensor and the trigger of every soil.



(b) Matric potential on a 5.85 ha field near Zaria, from mapped retention curves.



(c) Daisy workbench, step 3 of 9: ready-made experiments (trigger and sensor depth, dose, method) and the user's own.

Figure 6: The analysis in SWAFL Desktop 1.1.0 (real screenshots of the release).

6. Discussion

What the closed form adds to the van Genuchten curve. The van Genuchten–Mualem pair describes a soil at every suction; Eq. (4) says which suctions a cropped soil actually visits at the depth of a sensor. Its exponent, $2n + \ell(n-1)$, is the slope of the conductivity on a log–log scale at the dry end, so the reach is short where the conductivity collapses fast (large n) and long

where it fades slowly (small n), and α only rescales it. This complements the wet-end criteria of field capacity derived from the same parameters (Assouline and Or, 2014; Twarakavi et al., 2009): together they bracket the working range of a scheduled soil. That a relative and not an absolute conductivity sets the reach (Table 2) suggests that the limit is reached when the conductivity has fallen a fixed number of decades below its saturated value, which a field test can check.

What the fate of a millimetre adds to the pulsing question. The pulsed and intermittent drip simulations of Elmaloglou and Diamantopoulos (2007, 2008, 2009b) found that the application regime changes the losses below the root zone and that the effect depends on the soil. Across ten soils, ten doses, ten windows and three methods, the present data say how: the dose sets the drained share on a scale of one day of crop water use, the soil sets the plateau, the method sets how much of the water can be transpired at all, and the season sets how much the crop can use. Small doses do suppress drainage, but in a cropped field with an evaporating surface the water they save is lost to evaporation instead; the best compromise in these runs was a dose of about 2 mm. This is a marginal accounting and does not replace the event-scale wetting patterns of the earlier work, which a one-dimensional daily-mean analysis cannot resolve.

Practical rules. Three rules follow for sensor-scheduled irrigation. First, compute h_c from Eq. (4) for the soil and keep the trigger below $h_c/2$; on sands this means triggers of 50 to 100 cm, not the 300 to 600 cm of tables. Second, choose the dose against the daily crop water use: doses well below it evaporate, doses well above it drain at the soil's plateau. Third, overhead irrigation should be judged by its effect on transpiration, which can be nil, and not by the water delivered.

Limits and the experiments that would test the claims. The results are from one process model, one weather series (Taastrup, Denmark), one cereal rotation, one-dimensional columns, equilibrium Richards flow without hysteresis or water repellency (Diamantopoulos and Durner, 2013; Diamantopoulos et al., 2013; Elmaloglou and Diamantopoulos, 2009b), and effective van Genuchten parameters; the Mualem connectivity was not varied, the sensor was at 20 cm in E2 and E10, and the firing threshold of 5 mm a year is a choice (the cliffs are steep, so the intervals change little with it). Five of ten reaches are only bounded from below. The constant κ is a property of the model's uptake and should be measured, not assumed, in the field. The fates are 20-year means and do not resolve events. A field test would place matric-potential sensors on two coarse and two fine soils, schedule plots at triggers on both sides of h_c and at doses of 1, 4 and 15 mm, and measure drainage with lysimeters (Groh et al., 2020). The dynamic

non-equilibrium criterion Π_τ predicts when such a test must also resolve the lag of the sensor.

What is new, item by item. Table 3 sets each contribution of this article against what was established before it, so that what is taken from the literature and what is added here can be told apart.

Table 3: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Field capacity has intrinsic definitions from the hydraulic parameters (Assouline and Or, 2014; Twarakavi et al., 2009); uptake becomes limited by the conductance of the soil around the roots (Carminati and Javaux, 2020; Gardner, 1960).	A closed form for the dry-end counterpart, the *reach* of a suction trigger (Eq. (4)), with one constant shared by all soils.	Held-out soils within a factor of 2.3 (LOSO 0.18 decades); a threshold on absolute conductivity rejected (Table 2).
2	Triggers of 300 to 600 cm are tabulated for field crops (Allen et al., 1998).	These triggers lie beyond the reach of coarse soils: they never fire and the crop stays rainfed.	Sand and loamy sand at 300 cm: rainfed yield (Fig. 3d); a deeper sensor does not help (E3).
3	–	A reach-scaled trigger $h_\mathrm{t}/h_\mathrm{c}$ that places the irrigation responses of all soils on one axis.	Between-soil scatter falls from 54 % to 32 % of the total (Fig. 3b).
4	Pulsing, discharge and hydraulic properties change losses below the root zone, shown case by case (Elmaloglou and Diamantopoulos, 2007, 2008, 2009b).	The marginal fate of an irrigation millimetre across soils, doses, seasons and methods.	0.38 transpired, 0.25 soil evaporation, 0.19 other evaporation, 0.16 drained (Fig. 4).
5	–	A dose scale of the drained share that does not depend on the soil, and a soil-specific plateau.	$d_D = 4.1 \pm 0.6$ mm, $\rho = 0.07$ with n; plateau 0.10 to 0.45 (Fig. 5).

#	Established before	New in this article	Evidence here
6	Interception losses of sprinklers are known.	Overhead irrigation adds no transpiration: intercepted water replaces it.	$\eta_T = -0.08$ (all soils) and -0.01 (coarse loam).
7	Flow can be out of equilibrium with the retention curve (Diamantopoulos and Durner, 2012; Ross and Smettem, 2000).	A criterion, $\Pi_\tau = \tau / t_i$, for when this changes the firing of a trigger (proposed, not tested).	Derivation in the Supplementary Material.

7. Conclusions

A suction trigger has a reach, the largest suction a crop can impose at the sensor, and it follows from the van Genuchten–Mualem parameters by one closed form with one constant: $h_c = \alpha^{-1}(m^2/\kappa)^{1/[2n+\ell(n-1)]}$, $\kappa = 3.5 \times 10^{-10}$, valid out of sample within a factor of 2.3 over ten soils from sand to silty clay loam. Beyond the reach a trigger never fires; the common 300 cm trigger leaves sand and loamy sand rainfed. Each irrigation millimetre is transpired for 38 %, evaporated for 44 % and drained for 16 %; the drained share rises with the dose on a soil-independent scale of about 4 mm towards a soil-specific plateau, and overhead irrigation transpires nothing extra. Every run behind these numbers is open in `Data/Comprehensive_Summary`, the script reproduces them, and SWAFL Desktop lets any user apply them to a soil of their own.

Data, code and software availability

Software. SWAFL Desktop (version 1.1.0) is open source; the Software source is in the project repository (`https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Software`) and the program for Windows is SWAFL.exe (`https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/releases/latest/download/SWAFL.exe`). The whole workbench also runs in a browser, with nothing to install (link in the repository README).

Manual. The Manual (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/raw/main/Software/v1.1.0/SWAFL_Desktop/docs/manual_v1.1.0/SWAFL_Manual_v1.1.0.pdf) documents every control, screen and equation of the release, with labelled screenshots.

Data. The complete Daisy ensemble outputs (all 10 190 runs, raw and derived) are archived on Zenodo, doi: 10.5281/zenodo.23043626; the run table and a summary of every experiment are in the Data folder (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data).

Full simulation outputs. Every value of every log of every run (the annual water and nitrogen balances, the harvests and the crop state, year by year), one compressed table per experiment (`E00_Baselines` to `E16_Edge_Cases`), the one-row-per-run summary of all 10 190 runs (`00_summary_all_simulations`), the per-experiment summary (`00_SUMMARY_ALL_EXPERIMENTS`) and the script that made them (`summarize_results3.py`) are in the folder Data/Comprehensive_Summary of the repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data/Comprehensive_Summary); their 20-year means reproduce the run table within 2 %.

Supplementary Material. The Supplementary Material of this article gives the derivation of every equation, the mechanisms behind the results, the data of the analyses and the origin of every table and figure.

The data of every table and figure. One Excel workbook, Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx, holds the data of this work and its equations written as live Excel formulas connected to the data (sheets `Equations` and `EQ_*`): the run tables, the per-experiment summary, the derived results behind every table and figure, and a map that names, for each table and figure, the script that makes it, its input and its output. The map is also in `Data/Manuscript_Data/README.md`, and the workbook is rebuilt by `Data/Manuscript_Data/make_workbook.py`.

Code. Every check reported here has a script in `Manuscripts/reproducibility/p456` (Node.js scripts that load the shipped code or drive the shipped application, Python scripts for the analysis of the ensemble and the figures); `README_REPRODUCE.md` lists the steps. Daisy itself could not be run in the environment used for these checks, no field measurements are used, and the screenshots are archived rather than regenerated.

The analysis of this paper is `Manuscripts/reproducibility/analysis/p4_reach.py` (num-

bers in `Manuscripts/results/p4_numbers.json`); the concept figure is `p4_concept.py` and the data figure and table `data_overview.py`.

Author contributions

N.H.: conceptualisation, simulation, analysis, software, visualisation, writing.

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Competing interests

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A. Reach of every soil

Table 4: Observed reach interval (E2) and the predictions of Eq. (4) for the ten soils. Yields in Mg ha^{-1}.

ID	Soil	α (cm^{-1})	n	Observed (cm)	Eq. (4) (cm)	Yield 300 cm / rainfed / best
S03	Sand	0.145	2.68	100–150	199	1.13 / 1.13 / 3.04
S04	Loamy sand	0.124	2.28	200–250	426	1.38 / 1.38 / 4.54
S05	Sandy loam	0.075	1.89	800–1 000	1 618	4.35 / 1.85 / 6.02
S02	Coarse sand	0.0536	1.60	>10 000	5 373	6.25 / 1.63 / 6.32
S06	Loam	0.036	1.56	5 000–10 000	9 209	6.12 / 2.62 / 6.62
S08	Sandy clay loam	0.059	1.48	5 000–10 000	7 580	5.85 / 2.38 / 6.57
S07	Silt loam	0.020	1.41	>10 000	29 612	6.57 / 2.81 / 6.63
S01	Coarse loam	0.0961	1.353	>10 000	7 837	6.16 / 2.60 / 6.62
S09	Clay loam	0.019	1.31	>10 000	47 757	6.58 / 2.77 / 6.62
S10	Silty clay loam	0.010	1.23	>10 000	128 260	6.58 / 2.75 / 6.59

The two soils whose observed lower bound exceeds the prediction (coarse sand and coarse loam) are the two without a literature saturated conductivity in the set used here; their reach is underpredicted by at least a factor of 1.9 and 1.3, which a second constant (the power law of Table 2) only partly removes.

B. Reproducibility

Table 5: Scripts and inputs of every item. The data of every table and figure are also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`.

Item	Script	Input
Fig. 1	`p4_concept.py`	`p4_numbers.json`
Table 1	`data_overview.py`	`00_summary_all_simulations.csv.gz`
Figs. 2–5, Table 2, all numbers	`p4_reach.py`	`00_summary_all_simulations.csv.gz`
Fig. 6	screenshots of SWAFL Desktop 1.1.0	–